

Element H: Prototype Testing and Data Collection Plan

We tested:

- Thruster PWM response time (Target: $< 200\text{ms}$)
- Buoyancy net force at 3ft and 6ft (Target: $+0.25\text{N}$)
- Controller input delay (Target: $< 100\text{ms}$)
- Tether drag estimation (Measured: 1.4N at 6ft depth)

Each test was performed 5 times, charted, and photographed. Data showed thrusters were reliable, but tether weight skewed balance.

Future tests will simulate 10-minute operation cycles, float-release retention, and float-to-surface tracking. Data will be collected using a Gantt-charted system with timestamps.